Hesub Chaka

Columbia, MO | chakahesub@gmail.com | +1 571-279-9268 | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Data Science & Analytics graduate student with an emphasis in BioHealth Analytics and hands-on experience in research data management, data quality, and exploratory analysis. Skilled at converting complex, inconsistent datasets into reliable, structured information to support research and decision-making.

EDUCATION

University of Missouri–Columbia Columbia, MO
Master of Science in Data Science and Analytics. Emphasis: BioHealth Analytics. GPA: 3.83/4.0. *May 2027*
University of Missouri–Columbia Columbia, MO
Bachelor of Science in Information Technology. Certifications: Cybersecurity; Web Application Development; Digital Media Design.May 2025

SKILLS

Data & Analytics: Data Analysis, Statistical Analysis, Machine Learning, Exploratory Data Analysis, Data Visualization, Data Cleaning, Data Quality, Research Data Management
Programming & Querying: Python, SQL, R, C++, Java, TypeScript
Tools & Development: pandas, NumPy, Tableau, Jupyter, Excel, PostgreSQL, MongoDB, SQLite, Git, React, Node.js
Platforms: VS Code, Unix/Linux, GIS, Generative AI, AI-Assisted Analysis

EXPERIENCE

Paleontological Collections Data Assistant Mar 2026 – Present
University of Missouri–Columbia, Geology Department *Columbia, MO*
- Cataloged approximately 500 fossil specimens by converting historical records into structured Excel spreadsheets and a searchable museum database.
- Reviewed taxonomy, geology, locality, and specimen details against historical sources to improve record consistency and accuracy.
- Georeferenced collection records using GIS tools and corrected coordinates in the museum database.
- Documented discrepancies in historical records to support standardized, reliable collection data.

Research Assistant, MPOWER Research Project Mar 2026 – Aug 2026
University of Missouri–Columbia, Psychology Department *Columbia, MO*
- Cleaned and restructured research database records to improve consistency, organization, and accessibility.
- Validated research records and flagged discrepancies for correction, strengthening data reliability and documentation.

IT Operations Assistant May 2025 – Aug 2025
A.K.B.T General Construction *Addis Ababa, Ethiopia*
- Contributed to an employee commission-tracking application using React, Node.js, Express, and PostgreSQL to organize commission management.
- Troubleshot user issues, supported cloud-based tools, and maintained digital job-site reports, timesheets, and equipment logs.

Peer Learning Assistant Aug 2024 – May 2025
University of Missouri, College of Engineering *Columbia, MO*
- Guided students in UNIX and shell scripting through technical support and feedback.
- Reviewed and graded technical assignments, reinforcing programming and UNIX concepts.

Residential Advisor Aug 2023 – May 2026
University of Missouri, Residential Life *Columbia, MO*
- Supported approximately 300 residents through conflict mediation, resource referrals, and after-hours emergency response.
- Organized student programming and provided one-on-one support to strengthen community engagement.

PROJECTS

Backorder Prediction & Inventory Analysis Fall 2025
- Built machine learning workflows to predict product backorders from approximately 1.7 million inventory records and 22 predictors.
- Compared three pipelines combining anomaly detection, dimensionality reduction, and classification.
- Cleaned and encoded data, handled missing values, and prepared a balanced sample for model development.
- Used exploratory analysis and Tableau to visualize inventory, sales, forecasting, and backorder-risk patterns.

Chicago Crime Data Analysis & SQL Analytics Spring 2025
- Analyzed Chicago crime data in PostgreSQL to examine incidents, arrests, offense classifications, and geographic patterns.
- Wrote SQL queries using CTEs, joins, window functions, aggregations, and date-based analysis to identify trends.
- Created SQL views and time-series analyses, including five-year moving averages and district-level comparisons.
- Applied SQL regression analysis to compare gun-crime trends across police districts and citywide patterns.

TigerTraits, Student Matching Web Application 2024 – 2025

IT Capstone, University of Missouri–Columbia

- Developed a student-matching web application using personality and scenario-based responses.
- Built front-end and backend functionality with React, Node.js, and MongoDB.
- Collaborated in an Agile/Scrum team on development, research, analysis, and deliverables.